

NILESH PATIL

(+91) 9158244041 ◊ Dhule, IND
nileshbhilapatil123@gmail.com ◊ LinkedIn ◊ GitHub

SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, and Generative AI. Proficient in Python and Java, building RAG systems, LLM-powered applications, AI automation workflows, and deploying AI solutions. Strong foundation in data structures, algorithms, and problem-solving.

EDUCATION

B.Tech Artificial Intelligence and Machine Learning

2023 - 2027

R. C. Patel Institute of Technology, Shirpur, India

CGPA: 8.08

EXPERIENCE

AI Intern – Autom8x (A8X)

June 2026 – July 2026

- Developing Generative AI applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI automation techniques.
- Building AI workflows involving prompt engineering, API integration, data processing, and model evaluation.

SKILLS

- **Programming:** Python, Java
- **Machine Learning:** Scikit-learn, Feature Engineering, Model Training, Hyperparameter Tuning, Model Evaluation
- **Deep Learning & NLP:** TensorFlow, Keras, Neural Networks, NLP, Text Processing
- **Generative AI:** RAG, LangChain, LLM Integration, Fine-Tuning, Ollama, Hugging Face
- **AI Development:** FastAPI, Streamlit, REST APIs, AI Workflow Automation
- **MLOps & Cloud:** Docker, CI/CD, DVC, AWS
- **Databases:** MySQL, MongoDB, Vector Databases (FAISS)
- **Data & Tools:** NumPy, Pandas, Git, GitHub, Jupyter
- **Core CS:** Data Structures And Algorithms, OOPS

PROJECTS

SpaceAI RAG System – Astronomy QA Chatbot

2026

- Developed a Retrieval-Augmented Generation (RAG)-based chatbot capable of answering 100+ astronomy-related queries with enhanced contextual relevance.
- Processed 15+ astronomy eBooks into 8,000+ embeddings using FAISS with optimized top-k retrieval (k=5).
- Integrated Llama 3.2 via Ollama for accurate and context-aware response generation.
- Built and deployed REST APIs using FastAPI and Streamlit.
- **GitHub:** SpaceAI RAG System

Network Security ML Pipeline – Threat Detection System

2025

- Developed an end-to-end ML pipeline to classify network traffic as *Normal* or *Threat*.
- Automated data ingestion, preprocessing, model training, and evaluation.
- Achieved 94% accuracy with optimized model performance using precision, recall, and F1-score evaluation.
- Built bulk prediction API using FastAPI for real-time CSV analysis.
- Designed scalable pipeline workflows and deployed the system using Docker.
- Used AWS S3 and MongoDB for scalable storage and data management.
- **Live Demo:** Deployed Application

ACHIEVEMENTS

- Completed over **600+** Data Structures and Algorithms problems on LeetCode link.
- **Secured 2nd place** in the DSA Olympiad at SVKM, Dhule.